

RENEWABLE ENERGY AND GREEN HYDROGEN

POWER METRIC FIREWALL → ENERGY CATEGORY MAP → GRID INTEGRATION RAIL → PROJECT STATUS LADDER → ELECTROLYSIS PATHWAY → HYDROGEN CLAIM STACK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00POWER METRIC FIREWALL

POWER METRIC FIREWALLINSTALLED CAPACITYNAMEPLATE POWER CAPABILITYELECTRICAL ENERGY PRODUCED OVER TIMECAPACITY SHAREPORTFOLIO CAPABILITY, NOT OUTPUT SHARERELIABLE SUPPLYTIMING, FLEXIBILITY AND NETWORK MATTER

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
INSTALLED CAPACITY → nameplate power capability	CAPACITY SHARE → portfolio capability, not output share	RULE → never convert megawatts into generation without evidence
GENERATION → electrical energy produced over time	RELIABLE SUPPLY → timing, flexibility and network matter	INSTALLED CAPACITY → nameplate power capability

ECOLOGICAL UNIT / PARAMETER

INSTALLED CAPACITY → nameplate power capability
GENERATION → electrical energy produced over time

DECISIVE DISTINCTION

CAPACITY SHARE → portfolio capability, not output share
RELIABLE SUPPLY → timing, flexibility and network matter

SCALE / STATUS / EXCEPTION

RULE → never convert megawatts into generation without evidence
INSTALLED CAPACITY → nameplate power capability

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert megawatts into generation without evidence.

01

STAGE 01ENERGY CATEGORY MAP

ENERGY CATEGORY MAPSOLAR WIND HYDRO BIOMASS GEOTHERMALRENEWABLE SOURCES PLUS OTHER NON-FOSSIL SOURCESGENERATED ENERGY OUTPUTMANUFACTURED ENERGY CARRIERSOURCE, OUTPUT AND CARRIER ARE DIFFERENT CATEGORIES

RENEWABLE → solar wind hydro biomass geothermal

NON-FOSSIL → renewable sources plus other non-fossil sources

ELECTRICITY → generated energy output

HYDROGEN → manufactured energy carrier

SYSTEM / LEVEL / DEFINITION	STRUCTURE-FUNCTION MECHANISM	SCALE / EVIDENCE LIMIT
RENEWABLE → solar wind hydro biomass geothermal NON-FOSSIL → renewable sources plus other non-fossil sources	ELECTRICITY → generated energy output HYDROGEN → manufactured energy carrier	RULE → source, output and carrier are different categories RENEWABLE → solar wind hydro biomass geothermal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → source, output and carrier are different categories.

02

STAGE 02GRID INTEGRATION RAIL

GRID INTEGRATION RAILVARIABLE GENERATIONFORECAST RESOURCE-DEPENDENT OUTPUTMOVE POWER FROM RESOURCE TO DEMANDFLEXIBLE DEMAND AND BALANCINGMATCH SYSTEM CONDITIONSSHIFT ENERGY AND PROVIDE SELECTED SERVICESRELIABILITY IS A SYSTEM OUTCOME

SYSTEM INPUT / STARTING CONDITION	ECOLOGICAL PROCESS / TRANSFER	OUTCOME / FEEDBACK / LIMIT
VARIABLE GENERATION → forecast resource-dependent output	FLEXIBLE DEMAND AND BALANCING → match system conditions	DELIVERY → reliability is a system outcome
TRANSMISSION → move power from resource to demand	STORAGE → shift energy and provide selected services	VARIABLE GENERATION → forecast resource-dependent output

SYSTEM INPUT / STARTING CONDITION

VARIABLE GENERATION → forecast resource-dependent output
TRANSMISSION → move power from resource to demand

ECOLOGICAL PROCESS / TRANSFER

FLEXIBLE DEMAND AND BALANCING → match system conditions
STORAGE → shift energy and provide selected services

OUTCOME / FEEDBACK / LIMIT

DELIVERY → reliability is a system outcome
VARIABLE GENERATION → forecast resource-dependent output

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE → shift energy and provide selected services.

03

STAGE 03PROJECT STATUS LADDER

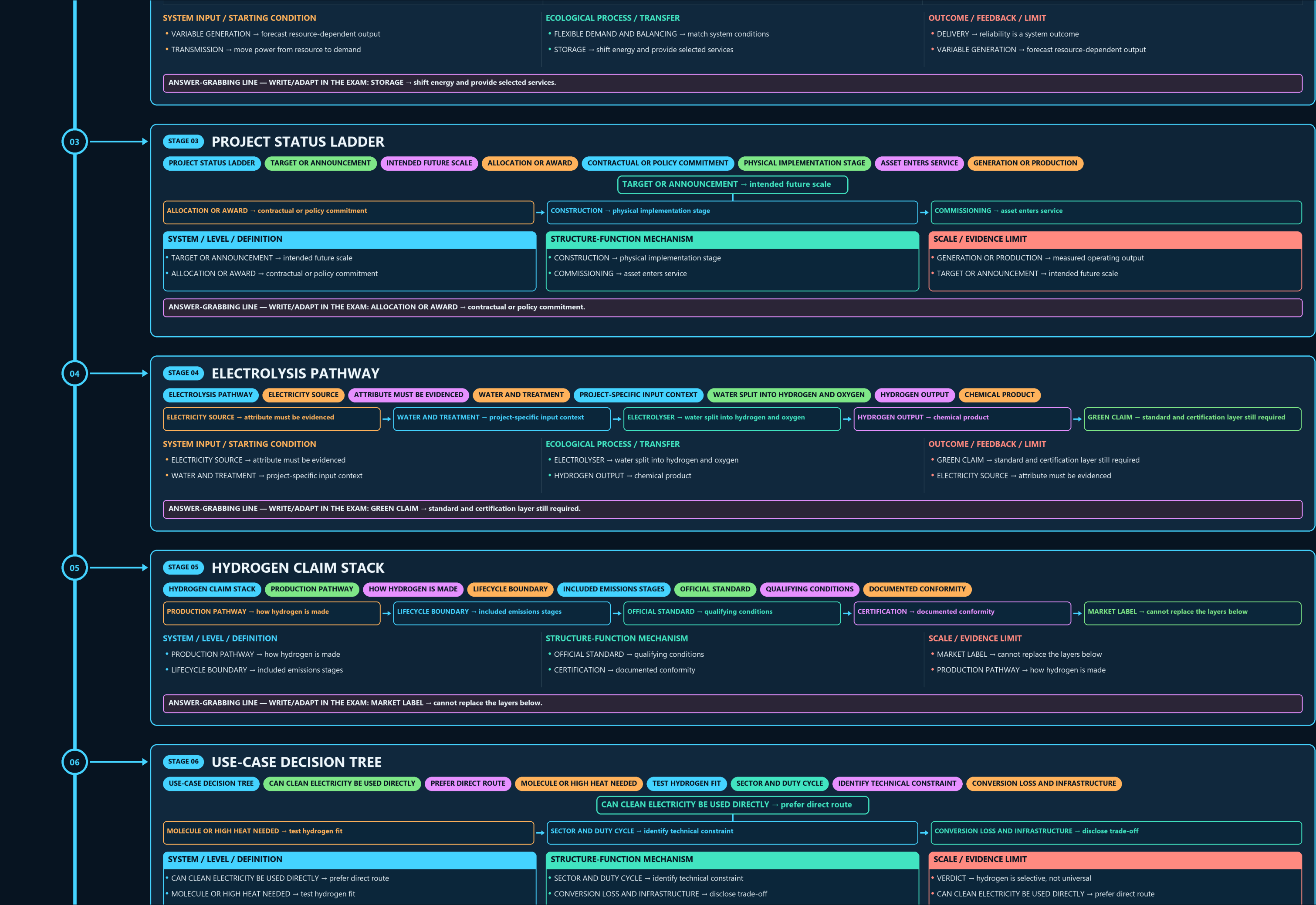
PROJECT STATUS LADDERTARGET OR ANNOUNCEMENTINTENDED FUTURE SCALEALLOCATION OR AWARDCONTRACTUAL OR POLICY COMMITMENTPHYSICAL IMPLEMENTATION STAGEASSET ENTERS SERVICEGENERATION OR PRODUCTION

TARGET OR ANNOUNCEMENT → intended future scale

ALLOCATION OR AWARD → contractual or policy commitment

CONSTRUCTION → physical implementation stage

COMMISSIONING → asset enters service



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MARKET LABEL → cannot replace the layers below.

06

STAGE 06 USE-CASE DECISION TREE

USE-CASE DECISION TREE CAN CLEAN ELECTRICITY BE USED DIRECTLY PREFER DIRECT ROUTE MOLECULE OR HIGH HEAT NEEDED TEST HYDROGEN FIT SECTOR AND DUTY CYCLE IDENTIFY TECHNICAL CONSTRAINT CONVERSION LOSS AND INFRASTRUCTURE

CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

SECTOR AND DUTY CYCLE → identify technical constraint

CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

SYSTEM / LEVEL / DEFINITION

- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route
- MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

STRUCTURE-FUNCTION MECHANISM

- SECTOR AND DUTY CYCLE → identify technical constraint
- CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

SCALE / EVIDENCE LIMIT

- VERDICT → hydrogen is selective, not universal
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR AND DUTY CYCLE → identify technical constraint.

07

STAGE 07 DERIVATIVE CONVERSION MAP

DERIVATIVE CONVERSION MAP BASE ENERGY CARRIER AND FEEDSTOCK NITROGEN-BASED DERIVATIVE METHANOL OR SYNTHETIC FUEL CARBON-CONTAINING DERIVATIVE CONVERSION AND DELIVERY EXTRA ENERGY AND INFRASTRUCTURE PRODUCT TARGETS AND OUTPUTS REMAIN SEPARATE

HYDROGEN → base energy carrier and feedstock

AMMONIA → nitrogen-based derivative

METHANOL OR SYNTHETIC FUEL → carbon-containing derivative

CONVERSION AND DELIVERY → extra energy and infrastructure

SYSTEM INPUT / STARTING CONDITION

- HYDROGEN → base energy carrier and feedstock
- AMMONIA → nitrogen-based derivative

ECOLOGICAL PROCESS / TRANSFER

- METHANOL OR SYNTHETIC FUEL → carbon-containing derivative
- CONVERSION AND DELIVERY → extra energy and infrastructure

OUTCOME / FEEDBACK / LIMIT

- RULE → product targets and outputs remain separate
- HYDROGEN → base energy carrier and feedstock

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → product targets and outputs remain separate.

08

STAGE 08 WATER AND SITE SCREEN

WATER AND SITE SCREEN FRESHWATER TREATED WATER OR OTHER VERIFIED SUPPLY PRETREATMENT REQUIREMENT BASIN AND LOCAL SCARCITY CONTEXT COMPETING DEMAND ECOLOGICAL AND SOCIAL CLAIMS PROJECT EVIDENCE, NOT A UNIVERSAL WATER NUMBER

SOURCE → freshwater treated water or other verified supply

QUALITY → pretreatment requirement

LOCATION → basin and local scarcity context

COMPETING DEMAND → ecological and social claims

SYSTEM / LEVEL / DEFINITION

- SOURCE → freshwater treated water or other verified supply
- QUALITY → pretreatment requirement

STRUCTURE-FUNCTION MECHANISM

- LOCATION → basin and local scarcity context
- COMPETING DEMAND → ecological and social claims

SCALE / EVIDENCE LIMIT

- VERDICT → project evidence, not a universal water number
- SOURCE → freshwater treated water or other verified supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → project evidence, not a universal water number.

09

STAGE 09 HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAIN HYDROGEN LEAVES ELECTROLYSER COMPRESSION LIQUEFACTION OR CONVERSION STORAGE AND TRANSPORT VESSEL PIPELINE TANKER OR PORT END USE EQUIPMENT AND SAFETY COMPATIBILITY DELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY

PRODUCTION → hydrogen leaves electrolyser

CONDITIONING → compression liquefaction or conversion

STORAGE AND TRANSPORT → vessel pipeline tanker or port

END USE → equipment and safety compatibility

OUTCOME → delivered service differs from production capacity

SYSTEM / LEVEL / DEFINITION

- PRODUCTION → hydrogen leaves electrolyser
- CONDITIONING → compression liquefaction or conversion

STRUCTURE-FUNCTION MECHANISM

- STORAGE AND TRANSPORT → vessel pipeline tanker or port
- END USE → equipment and safety compatibility

SCALE / EVIDENCE LIMIT

- OUTCOME → delivered service differs from production capacity
- PRODUCTION → hydrogen leaves electrolyser

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → delivered service differs from production capacity.

09

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → project evidence, not a universal water number.

STAGE 09

HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAINHYDROGEN LEAVES ELECTROLYSERCOMPRESSION LIQUEFACTION OR CONVERSIONSTORAGE AND TRANSPORTVESSEL PIPELINE TANKER OR PORTEND USEEQUIPMENT AND SAFETY COMPATIBILITYDELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY

PRODUCTION → hydrogen leaves electrolyser→CONDITIONING → compression liquefaction or conversion→STORAGE AND TRANSPORT → vessel pipeline tanker or port→END USE → equipment and safety compatibility→OUTCOME → delivered service differs from production capacity

SYSTEM / LEVEL / DEFINITION

• PRODUCTION → hydrogen leaves electrolyser

• CONDITIONING → compression liquefaction or conversion

STRUCTURE-FUNCTION MECHANISM

• STORAGE AND TRANSPORT → vessel pipeline tanker or port

• END USE → equipment and safety compatibility

SCALE / EVIDENCE LIMIT

• OUTCOME → delivered service differs from production capacity

• PRODUCTION → hydrogen leaves electrolyser

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTCOME → delivered service differs from production capacity.

10

STAGE 10

MISSION AND SIGHT MAP

MISSION AND SIGHT MAPMISSION TARGETINTENDED FUTURE PRODUCTION SCALESIGHT MANUFACTURINGELECTROLYSER CAPABILITY SUPPORTSIGHT PRODUCTIONHYDROGEN OUTPUT SUPPORTCOMMISSIONED OUTPUT

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
MISSION TARGET → intended future production scale	SIGHT PRODUCTION → hydrogen output support	CLIMATE OUTCOME → separate lifecycle result
SIGHT MANUFACTURING → electrolyser capability support	COMMISSIONED OUTPUT → later measured evidence	MISSION TARGET → intended future production scale

ECOLOGICAL UNIT / PARAMETER

• MISSION TARGET → intended future production scale

• SIGHT MANUFACTURING → electrolyser capability support

DECISIVE DISTINCTION

• SIGHT PRODUCTION → hydrogen output support

• COMMISSIONED OUTPUT → later measured evidence

SCALE / STATUS / EXCEPTION

• CLIMATE OUTCOME → separate lifecycle result

• MISSION TARGET → intended future production scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMMISSIONED OUTPUT → later measured evidence; CLIMATE OUTCOME → separate lifecycle result.

11

STAGE 11

ENERGY-TRANSITION ANSWER SPINE

ENERGY-TRANSITION ANSWER SPINECAPACITY GENERATION RELIABILITY SOURCE AND CARRIERGRID INTEGRATION ELECTROLYSIS CERTIFICATION AND DELIVERYDIRECT ELECTRIFICATION BEFORE SELECTIVE HYDROGEN USETARGET AWARD COMMISSION OUTPUT AND OUTCOME

DATED MNRE POWER-SYSTEM AND STANDARDS EVIDENCE

DEFINE → capacity generation reliability source and carrier→TRACE → grid integration electrolysis certification and delivery→SCREEN → direct electrification before selective hydrogen use→SEPARATE → target award commission output and outcome→VERIFY → dated MNRE power-system and standards evidence

SYSTEM / LEVEL / DEFINITION

• DEFINE → capacity generation reliability source and carrier

• TRACE → grid integration electrolysis certification and delivery

STRUCTURE-FUNCTION MECHANISM

• SCREEN → direct electrification before selective hydrogen use

• SEPARATE → target award commission output and outcome

SCALE / EVIDENCE LIMIT

• VERIFY → dated MNRE power-system and standards evidence

• DEFINE → capacity generation reliability source and carrier

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated MNRE power-system and standards evidence.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENTLARGE-SCALE SOLAR AND WIND INFRASTRUCTURE REQUIRES SUBSTANTIAL LAND AREABALANCED FRAMINGTHIS IS NOT A CASE WHERE EITHER RENEWABLE ENERGYBATTERY-STORAGE COSTS HAVE BEEN FALLING BUT REMAIN A SUBSTANTIALANALYTICAL POINT

INDIA'S RENEWABLE-CAPACITY-ADDITION TARGETS (500 GW NON-FOSSIL BYGREEN-HYDROGEN PRODUCTION-COST FIGURES AND ELECTROLYSER-CAPACITY FIGURES CHANGE RAPIDLY

OPTIONAL ECOLOGICAL PROCESS DEPTH

• CAUTION: Large-scale solar and wind infrastructure requires substantial land area and, for wind

• CAUTION: Balanced framing: this is not a case where either renewable energy or species

• CAUTION: Battery-storage costs have been falling but remain a substantial cost component of a

• CAUTION: Analytical point: India's renewable-capacity-addition targets (500 GW non-fossil by

SCALE, PARAMETER OR STATUS LIMIT

• CAUTION: Green-hydrogen production-cost figures and electrolyser-capacity figures change rapidly

• CAUTION: Progress toward the National Green Hydrogen Mission's 5 MMT/year 2030 target and the

• CAUTION: Quantifying the precise trade-off between renewable-infrastructure expansion and

• Solar and wind generation are variable/non-dispatchable, requiring storage and grid

COMPARATIVE RESTORATION NUANCE

• Green hydrogen's cost-competitiveness depends on both falling electrolyser/renewable

• The Supreme Court has engaged with the tension between renewable-energy transmission

• The SIGHT Programme's dual incentive structure (electrolyser manufacturing plus

• NOT: India's renewable-energy challenge today is primarily about building enough generation

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels

RENEWABLE ENERGY AND GREEN HYDROGEN • TILE 4/4 • same master rows 7420-10654 of 10654 • end